LAYER EDGE: Model → Silicon · Model → Stack · Silicon → Model · Cross-layer loop · Claimed · Organization

MECHANISM: Data · Feedback & verification · Search & RL · Agents · Cross-layer

37 entries in catalog

H EXECUTOR: PEOPLE 35

- **FlashAttention** — IO-aware attention
- **NSA** — HW-aligned sparse attention
- **DeepSeek-V3.1** — FP8 for a future chip
- **DeepGEMM** — hand-tuned FP8 kernels
- **MTIA v2** — workload-driven ASIC
- **Pangu Ultra** — built for Ascend NPUs

HW-aware NAS · Low precision · MoE & sparsity

L0 GENERATES ONCE 19

- **ChipNeMo** — domain-adapted LLM
- **Chip-Chat** — chat-driven tape-out
- **VeriGen** — Verilog code LLM
- **RTLCoder** — open synthetic RTL data
- **GPT4AIGChip** — HLS for AI accelerators
- **LLM Compiler** — IR-level optimization

Domain pretraining · SFT · Synthetic data · RAG

L1 CLOSED LOOP, GIVEN EVALUATOR 37

MODEL → SILICON

- **AutoChip** — compile & sim feedback
- **RTLFixer** — ReAct debug loop
- **QiMeng-CodeV-R1** — equivalence reward
- **Self-Evolved ABC** — evolves synthesis tool
- **AlphaChip** — RL macro placement
- **PrefixRL** — RL adders in Hopper

Testbench reward · Equivalence check · RLVR · Multi-agent

MODEL → STACK

- **CUDA Agent** — agentic RL for kernels
- **GEAK** — Triton kernel agent
- **KernelBlaster** — memory across GPUs
- **daVinci-kernel** — verified skill library
- **AlphaTensor** — HW-timed matmul search
- **AlphaDev** — sorting merged in LLVM

Profiler loop · Test-time search · Skill library · Anti-hacking

L2 SPEC IN · SYSTEM OUT 4

- **Design Conductor** — spec → RISC-V core in 12 h
- **ASIC-Agent** — RTL → OpenLane → Caravel
- **QiMeng** — HW + SW design agents
- **GEAK v4** — whole serving workloads
- **Qwen chip–model co-evo** — NoC spec → layout · Day-0 deploy
- **EDA vendor agents** — Cadence · Synopsys · Siemens

Spec → GDSII · Task decomposition · End-to-end profiling · Multi-backend

L3 AI PICKS THE LAYER 2

- **Design Conductor 2.0** — algorithm paper → accelerator
- **HSCO-Bench** — SW needs → SoC accelerator
- *Model × kernel × RTL agent* — open: no instance yet

Cross-layer gain · Joint J(M,S,H) · World model · Cross-layer env

L4 FEEDS THE DESIGNER 1 + 2

- **AlphaEvolve** — TPU circuit + Gemini kernel
- **Jalapeño** — model-aided chip · undisclosed
- **OpenAI × Broadcom** — ~20 optimizations · claimed

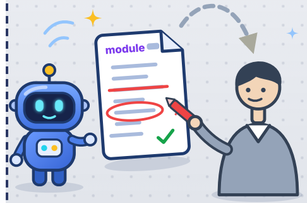
Kernel → training · Circuit → next TPU · Human sign-off

Human-Led Co-Design



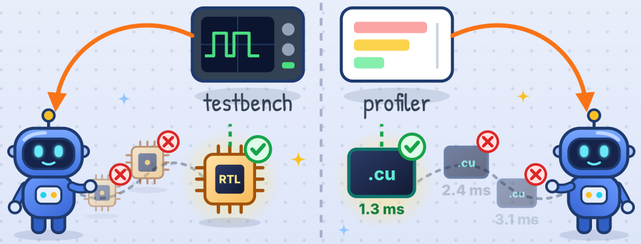
Design Assistance

drafts once · human iterates



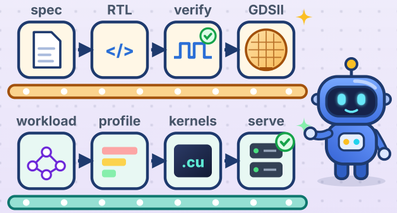
Local Autonomous Optimization

propose · test · keep



End-to-End Autonomous Design

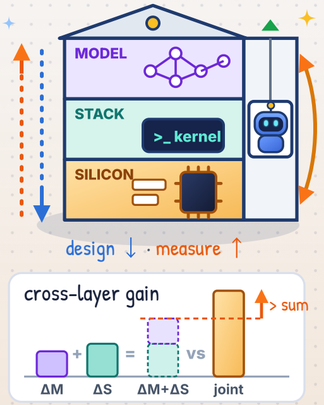
spec in → system out · 12 h



no hand-offs between stages

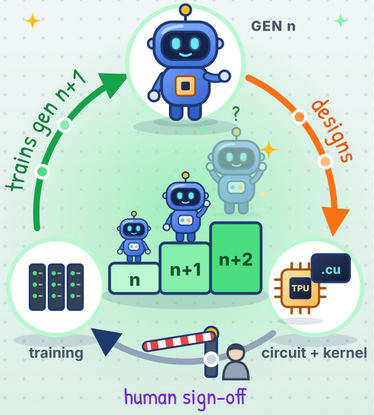
Cross-Layer Co-Design

which floor to change?



Recursive Self-Enhancement

the designer improves itself



H Human · L0 Assist · L1 Optimize · L2 End-to-End · L3 Cross-Layer · L4 Recursive

L5 self-expansion · L6 open-ended evolution →